

Adam Williams

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Profile

Staff software engineer specializing in Go backend systems, building and owning distributed services at scale. Biased toward simple, durable code.

At Fastly, works across both sides of the platform: the edge points of presence (POPs) and the Kubernetes-based control plane, primarily in Go and Rust.

Experience

Fastly, Staff Software Engineer

Apr 2021 – present

- Design and own the Go backend for the Secret Store product, including its public API, deployed on Kubernetes.
- Lead the migration of the Config Store API out of a legacy monolith into a standalone Go service on Kubernetes, so it scales and ships features independently.
- Build the observability and monitoring that verified a safe migration of a Go configuration delivery system from eager to on-demand delivery across thousands of edge nodes, reclaiming the disk capacity needed to keep onboarding customers.
- Migrate on-the-fly video packaging customers onto a new delivery configuration layer (VCL), letting Fastly ship packaging changes without customer involvement; support and extend the underlying Go service.
- Unblock the GA release of ACLs on Compute by implementing hostcalls and configuration delivery; contribute further Compute platform features in Rust.
- Add Secret Store and Config Store commands to the Fastly CLI, the official tool customers use to deploy and configure services.

Comcast, Senior Software Engineer (Engineer 4)

Feb 2015 – Apr 2021

- Designed an MPEG-TS parsing library in Go and built a video stream monitoring application on it, reporting metrics with Prometheus and deployed with Docker and Kubernetes; added an automated testing framework that improved accuracy and development speed.
- Built a Go decoder for set-top box commands from protocol specifications.
- Deployed services nationally across multiple data centers with Kubernetes and Helm; developed a custom CNI plugin.
- Built a network monitoring tool that uncovered a long-standing production DNS issue, then resolved it.
- Designed and built the native Go client for Apache Pulsar alongside the project's core maintainers; released as open source at github.com/Comcast/pulsar-client-go.
- Selected for the Open Source Fellowship Program; received a top contributor award.

Cabify, Senior Developer

Oct 2013 – Feb 2015

- Owned infrastructure for a service operating in four countries; provisioned and maintained servers on AWS and Linode with Ansible, and monitored them with InfluxDB.
- Delivered product features across Ruby on Rails, Node.js, and Go services, and built internal developer tooling.
- Introduced Go to the company, replacing Node.js services and improving reliability.

Education

BS University of Minnesota, BS in Mathematics – Minneapolis, MN

BA University of Minnesota, BA in Spanish Studies – Minneapolis, MN

Projects

wifi-presence

Open-source OpenWrt package providing presence detection based on Wi-Fi client connections.

Skills

Languages: Go (primary), Rust, Ruby

Infrastructure: Kubernetes, Helm, Docker, Prometheus, AWS, GCP, Ansible

Domains: Distributed systems, API backends, network protocols, video streaming, observability